



A. Core Concepts (VERY IMPORTANT)

1. Network Security vs Data vs Cyber

Type	Meaning
Network Security	Protects data during transmission (e.g., firewalls, encryption)
Data Security	Protects stored data (e.g., encryption at rest)
Cyber Security	Broad field (network + data + systems + users)

👉 Trick Answer:

Cyber Security = Umbrella term

2. Substitution vs Transposition

Type	Idea
Substitution Cipher	Replace letters
Transposition Cipher	Rearrange letters

👉 Examples:

- Caesar → Substitution
- Vigenère → Substitution
- Playfair → Substitution (digraph)
- Rail Fence → Transposition

👉 Trick Question:

Playfair is NOT transposition (many students say wrong)



B. Algorithm-wise Viva Notes

◆ 1. Caesar Cipher

Math:

$$C = (P + k) \bmod 26$$

Key Idea:

- Fixed shift

Weakness:

- Easy brute force (25 keys)

Application:

- Not used in real security
-

◆ 2. Vigenère Cipher

Math:

$$C_i = (P_i + K_i) \bmod 26$$

Key Idea:

- Multiple shifts (polyalphabetic)

Upgrade over Caesar:

- More secure (no fixed pattern)

Weakness:

- Repeating key → frequency attack

👉 Trick Question:

Why stronger than Caesar?

✓ Because key changes per letter

◆ 3. Playfair Cipher

Key Idea:

- Encrypts pairs (digraph)

Important Rules:

- Same row → shift right
- Same column → shift down
- Rectangle → swap columns

👉 **Trick Question:**

Why use digraph?

✓ To reduce frequency attack

◆ 4. Rail Fence Cipher

Type:

- Transposition

Idea:

- Zig-zag pattern

👉 **Trick Question:**

Does it change letters?

✗ No → only rearranges

🔒 C. Modern Cryptography

◆ 5. Diffie-Hellman

Purpose:

- Key exchange

Key Idea:

- No key is transmitted directly

👉 **Trick Question:**

Does it encrypt data?

✗ No → only shares key

◆ 6. RSA

Purpose:

- Encryption + Digital Signature

Key Idea:

- Public key + Private key

👉 **Difference from Diffie-Hellman:**

Feature	Diffie-Hellman	RSA
Use	Key exchange	Encryption
Key type	Shared secret	Public-private
Encryption	No	Yes



D. Code-Based Questions (VERY COMMON)

They may ask:

👉 “What is happening in your code?”

You should say:

1. Convert plaintext → numbers
 2. Apply formula
 3. Use modulo 26
 4. Convert back to letters
-

Example (Vigenère):

ATTACK + LEMON

P	A	T	T	A	C	K
K	L	E	M	O	N	L

$$C = (P + K) \bmod 26$$

👉 Output: LXFOPV



E. MOST IMPORTANT QUESTION

! “Where will you use this?”

You MUST answer like this:

Algorithm	Real-life Use
Caesar	Learning only
Vigenère	Historical use
Playfair	Military (old)
Rail Fence	Puzzle/learning
Diffie-Hellman	Secure key exchange (HTTPS)
RSA	Banking, SSL, Digital signatures

👉 Golden Line:

Classical ciphers are for learning, modern cryptography is used in real systems.

! F. Common Trap Questions

👉 1. Is Vigenère secure today?

❌ No

👉 2. Is RSA symmetric?

❌ No (asymmetric)

👉 3. Is Diffie-Hellman encryption?

❌ No

👉 4. Which is stronger: Caesar or Vigenère?

✓ Vigenère

👉 5. What is brute force?

✓ Trying all keys

🧠 G. How to Impress Examiner

Say these naturally:

- “This is a substitution cipher because characters are replaced.”
- “Modulo operation ensures wrap-around within alphabet.”
- “Security depends on key length.”
- “Modern systems use asymmetric cryptography.”



1. VERY FIRST QUESTIONS (Almost Guaranteed)

? What is Cryptography?

👉 Answer:

Cryptography is the technique of securing communication by converting plaintext into ciphertext using algorithms and keys.

? Goals of Cryptography?

👉 Use **CIA + 2 more**:

- Confidentiality
 - Integrity
 - Authentication
 - Non-repudiation
 - Availability
-

? What is Network Security?

👉 Protects data **during transmission**

? Data vs Network vs Cyber Security

Type	Focus
Data Security	Stored data
Network Security	Data in transit
Cyber Security	Everything

👉 **Golden Line:**

Cyber Security = umbrella term

2. CLASSIFICATION (VERY IMPORTANT)

? Types of Cryptography

1. Symmetric

- Same key
- Fast
- Example: Caesar, Vigenère, Hill

2. Asymmetric

- Public + Private key
 - Example: RSA
-

? Cipher Types

Type	Meaning
Substitution	Replace letters
Transposition	Rearrange letters

3. EXPERIMENT-WISE VIVA (WITH TRICKS)

◆ Experiment 1: Caesar Cipher

$$C = (P + k) \bmod 26$$

👉 Type: Substitution

👉 Key Idea: Fixed shift

👉 Weakness:

- Only 25 keys → brute force easy

👉 Viva Trap:

? Is it secure?

✗ No

◆ Experiment 2: Playfair Cipher

👉 Type: Substitution (Digraph)

👉 Key Idea:

- Encrypts **pairs**

👉 Why better than Caesar?

✓ Reduces frequency patterns

👉 Trick:

? Is it transposition?

✗ No

◆ Experiment 3: Hill Cipher

$$C = K \times P \bmod 26$$

👉 Type: Polygraphic substitution

👉 Based on:

✓ Linear Algebra (Matrix)

👉 Trick Questions:

? Why key must be invertible?

✓ For decryption

? If determinant has no inverse?

✗ Decryption impossible

◆ Experiment 4: Vigenère Cipher

$$C_i = (P_i + K_i) \bmod 26$$

👉 Type: Polyalphabetic substitution

👉 Upgrade from Caesar:

✓ Different shift per letter

👉 Weakness:

- Repeating key → pattern

👉 Trick:

? Stronger than Caesar?

✓ Yes

◆ Experiment 5: Rail Fence Cipher

👉 Type: Transposition

👉 Idea:

- Zigzag pattern

👉 Trick:

? Does it change letters?

✗ No

◆ Experiment 6: RSA Algorithm

$$C = P^e \bmod n$$

👉 Type: Asymmetric

👉 Key Idea:

- Public + Private key

👉 Important:

- Based on **prime factorization difficulty**

👉 Trick:

? Is RSA symmetric?

✗ No

◆ Experiment 7: Diffie-Hellman

👉 Type: Key Exchange

👉 Key Idea:

- Share secret without sending it

👉 Trick:
? Is it encryption?
❌ No

🔪 4. VERY IMPORTANT COMPARISONS

🔥 Caesar vs Vigenère

Feature	Caesar	Vigenère
Key	Single	Multiple
Security	Weak	Stronger

🔥 RSA vs Diffie-Hellman

Feature	RSA	Diffie
Use	Encryption	Key exchange
Key	Public/Private	Shared
Encrypt data	Yes	No

🔥 Substitution vs Transposition

Feature	Substitution	Transposition
Change letters	Yes	No
Example	Caesar	Rail Fence

💻 5. CODE-BASED VIVA (VERY COMMON)

? What happens in your code?

👉 Say confidently:

- Convert letters → numbers
- Apply formula

- Use modulo 26
- Convert back

? Why modulo 26?

👉 Answer:

Because English alphabet has 26 letters (wrap-around)

🎯 6. MOST REPEATED QUESTION

! “Where will you use this?”

Algorithm	Use
Caesar	Learning
Playfair	Old military
Hill	Theoretical
Vigenère	Historical
Rail Fence	Puzzle
RSA	Banking, SSL
Diffie-Hellman	Key exchange (HTTPS)

👉 **Best Answer Line:**

Classical ciphers are mainly for learning; modern cryptography like RSA is used in real-world systems.

⚠️ 7. HIGH-LEVEL TRICK QUESTIONS

👉 ? What is brute force?

✓ Trying all keys

👉 ? What is key space?

✓ Total possible keys

👉 ? What makes encryption strong?

✓ Key length + randomness

- 👉 ? Why classical ciphers fail?
 - ✓ Pattern detection
-

8. HOW TO ANSWER SMARTLY

Always connect:

👉 Concept → Math → Code → Real Life

Example:

“This is a substitution cipher because characters are replaced. The formula uses modulo 26 for alphabet wrapping, and in code we convert characters to ASCII values. However, it is not secure for real-world use.”

FINAL ADVICE (VERY IMPORTANT)

Your teacher is checking:

- ✓ Do you **understand math**
- ✓ Can you **explain code**
- ✓ Can you **apply in real life**

SECTION 1: BASIC CONCEPTS (VERY IMPORTANT)

1. What is cryptography?

👉 Technique to secure communication using encryption.

2. What is plaintext?

👉 Original readable message.

3. What is ciphertext?

👉 Encrypted unreadable message.

4. What is encryption?

👉 Converting plaintext → ciphertext.

5. What is decryption?

👉 Converting ciphertext → plaintext.

6. What is a key?

👉 Parameter used in encryption/decryption.

7. What is an algorithm?

👉 Step-by-step process for encryption.

8. Goals of cryptography?

👉 Confidentiality, Integrity, Authentication, Non-repudiation.

9. What is brute force attack?

👉 Trying all possible keys.

10. What is key space?

👉 Total number of possible keys.



SECTION 2: TYPES OF CRYPTOGRAPHY

11. What is symmetric encryption?

👉 Same key for encryption & decryption.

12. What is asymmetric encryption?

👉 Uses public and private keys.

13. Difference between symmetric & asymmetric?

Symmetric	Asymmetric
One key	Two keys
Fast	Slow

14. Example of symmetric?

👉 Caesar, Vigenère, Hill

15. Example of asymmetric?

👉 RSA



SECTION 3: CIPHER TYPES

16. What is substitution cipher?

👉 Replaces characters.

17. What is transposition cipher?

👉 Rearranges characters.

18. Example of substitution?

👉 Caesar

19. Example of transposition?

👉 Rail Fence



SECTION 4: CAESAR CIPHER

20. Formula?

$$C = (P + k) \bmod 26$$

21. Why mod 26?

👉 26 letters in alphabet.

22. Weakness?

👉 Only 25 keys → easy brute force.

23. Type?

👉 Substitution cipher.

24. Is it secure?

👉 No.



SECTION 5: PLAYFAIR CIPHER

25. Type?

👉 Digraph substitution.

26. Why use pairs?

👉 Reduces frequency attack.

27. Matrix size?

👉 5×5

28. Why I and J combined?

👉 Only 25 slots.

29. Rule if same row?

👉 Shift right.

30. Rule if same column?

👉 Shift down.

SECTION 6: HILL CIPHER

31. Based on?

👉 Linear algebra.

32. Formula?

$$C = K \times P \text{ mod } 26$$

33. Why matrix must be invertible?

👉 Needed for decryption.

34. What if not invertible?

👉 Cannot decrypt.

35. Type?

👉 Polygraphic substitution.

SECTION 7: VIGENÈRE CIPHER

36. Type?

👉 Polyalphabetic substitution.

37. Formula?

$$C_i = (P_i + K_i) \text{ mod } 26$$

38. Advantage over Caesar?

👉 Multiple shifts.

39. Weakness?

👉 Repeating key pattern.



SECTION 8: RAIL FENCE

40. Type?

👉 Transposition cipher.

41. Does it change letters?

👉 No.

42. Pattern?

👉 Zigzag.



SECTION 9: RSA

43. Type?

👉 Asymmetric.

44. Keys?

👉 Public + Private.

45. Formula?

$$C = P^e \bmod n$$

46. Based on?

👉 Prime factorization difficulty.

47. Why secure?

👉 Hard to factor large numbers.

SECTION 10: DIFFIE-HELLMAN

48. Purpose?

👉 Key exchange.

49. Does it encrypt?

👉 No.

50. Main idea?

👉 Shared secret key.

SECTION 11: COMPARISON

51. Caesar vs Vigenère?

👉 Single vs multiple shift.

52. RSA vs Diffie?

👉 Encryption vs key exchange.

SECTION 12: CODE VIVA

53. Why use ord()?

👉 Convert char → number.

54. Why chr()?

👉 Convert number → char.

55. Why modulo?

👉 Wrap around alphabet.

! SECTION 13: TRICK QUESTIONS

56. Is Rail Fence secure?

👉 No.

57. Is Playfair modern?

👉 No.

58. Can Caesar resist brute force?

👉 No.

🔥 SECTION 14: ADVANCED

59. What is frequency analysis?

👉 Studying letter frequency.

60. Which cipher resists it?

👉 Vigenère (partially).

61. What is modular inverse?

👉 Number that satisfies:

$$(a \times x) \bmod m = 1$$

62. What is GCD?

👉 Greatest Common Divisor.

🎯 SECTION 15: REAL-LIFE APPLICATION

63. Where is RSA used?

👉 HTTPS, banking.

64. Diffie-Hellman use?

👉 Secure key exchange.

SECTION 16: SUPER FAST FIRE (IMPORTANT)

65. Cipher vs code?

👉 Cipher = algorithm, Code = mapping

66. Key length importance?

👉 Longer = more secure

67. Attack types?

👉 Brute force, frequency

FINAL QUICK REVISION (MOST IMPORTANT 20)

👉 Memorize these:

- Cryptography definition
 - CIA triad
 - Symmetric vs Asymmetric
 - Substitution vs Transposition
 - Caesar formula
 - Hill matrix invertible
 - Vigenère key repeat
 - Rail Fence zigzag
 - RSA public/private
 - Diffie key exchange
-